Lecture Notes
EC226 Term 1 2025/26

Limited Dependent Variable (LDV) Models

Reading:

- Stock and Watson (2003): Chapter 11
- Dougherty (2016): Chapter 10
- Wooldridge (2013): Chapter 17

Contents

1 Introduction

In these models we have, for individual i, the dependent variable y_i takes on two values: 0 and 1. These are called Dummy Dependent Variable Models and/or Qualitative Choice Models. Examples:

1 Vote or NOT

2 Go to University or NOT

3 Buy a car or NOT

Where we might have, e.g. $y_i = 1$ if voted and $\mathrm{y_i} = 0$ if not vote and we are interested in the factors (i.e. individual i's characteristics) which influence the decision to vote.
In these model we assume there is some unobserved latent variable, y_i^*, which measures individual i's desired satisfaction (or net utility) from undertaking the activity and we believe this is related to some explanatory variables, $x_{1i}, x_{2i}, x_{3i}, \ldots x_{ki}$. In particular we believe this net utility is a linear function of these variables, such that:

$$y_i^* = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots + \beta_k x_{ki} + \varepsilon_i \tag{1}$$

where ε_i is the usual unexplained component (error term). For shorthand we can write as:

$$y_i^* = {x_i}'\beta + \varepsilon_i$$

where ${x_i}' = (1, x_{1i}, x_{2i}, \ldots, x_{ki})$ and $\beta' = (\beta_0, \beta_1, \ldots, \beta_k)$. Clearly if y_i^* were observed we would apply OLS to equation (1). We relate the latent variable, y_i^*, to the observed variable y_i as follows:[1]
$y_i = 1$ if $y_i^* \geq 0$ and $= 0$ if $y_i^* < 0$ where $y_i = 1$ with probability, p_i, and $y_i = 0$ with probability, $1 - p_i$. As this is a Bernoulli trial:

$$E(y_i) = 0.(1 - p_i) + 1.p_i = p_i = P(y_i = 1)$$

We normally refer to a coefficient as measuring the change in the expected value of y for a unit increase in some explanatory variable, x_j, in this case we are measuring the change in the $P(y_i = 1)$ for a unit increase in some explanatory variable, x_j.
Now let us look more carefully at $P(y_i = 1)$:

$$E(y_i) = P(y_i = 1) = P(y_i^* > 0) = P(\varepsilon_i > -{x_i}'\beta) = P(\varepsilon_i < {x_i}'\beta) = F({x_i}'\beta) \tag{2}$$

where, $F({x_i}'\beta)$= Cumulative distribution function (cdf) and is the probability of observing a value less than ${x_i}'\beta$. Now y_i is defined as:

$$y_i = E(y_i) + u_i \tag{3}$$

There are three alternative models for defining the cdf $F({x_i}'\beta)$, these are:

1. the linear probability model (LPM)
2. the logit model
3. the probit model

[1] It is arbitrary what we choose as the cut-off as a non-zero cut-off will simply be picked up by or estimate b_0

2 Alternative binary choice models

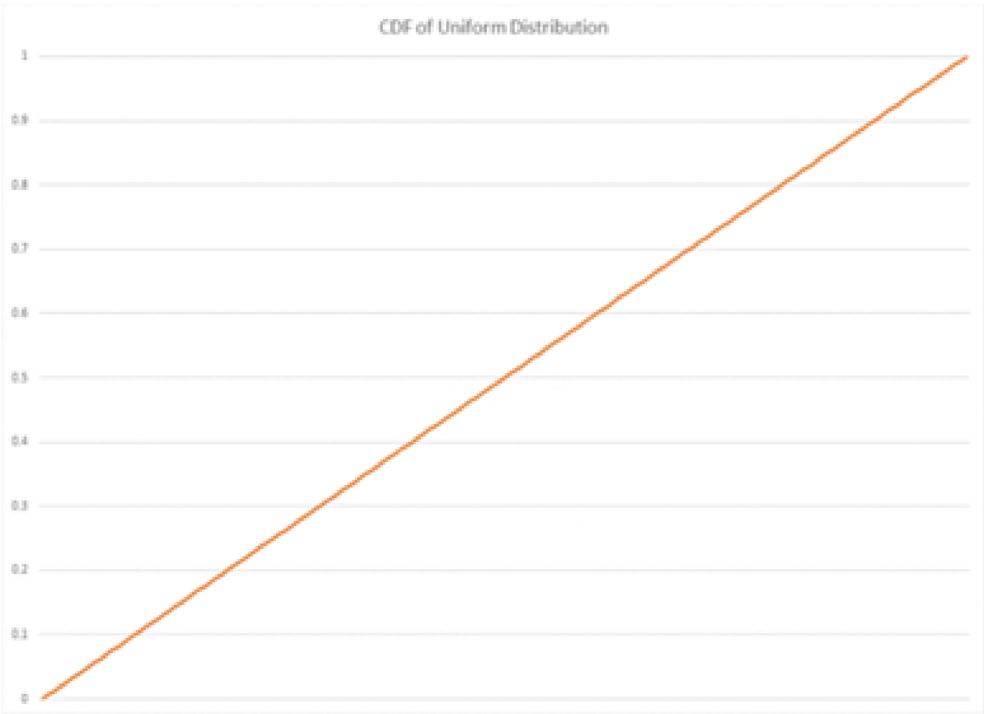
2.1 Linear Probability Model (LPM)

This has the cdf written as

$$E(y_i) = F(x_i{}'\beta) = U(x_i{}'\beta) = x_i{}'\beta$$

where U is a uniform distribution, which implies a CDF as seen in Figure 1, in which case we have $y_i = E(y_i) + u_i$, and therefore $y_i = x_i{}'\beta + u_i$.

Figure 1:



There are three problems with this model:

1. Non-normality of the error term:

$$y_i = 1 \Rightarrow u_i = 1 - x_i{}'\beta$$

$$y_i = 0 \Rightarrow u_i = -x_i{}'\beta$$

and therefore u_i only takes on 2 values and therefore cannot be normal.

2. Heteroscedasticity of the error term:

$$E(u_i) = \Pr[y_i = 1](1 - x_i{}'\beta) + \Pr[y_i = 0](-x_i{}'\beta) = x_i{}'\beta(1 - x_i{}'\beta) + (1 - x_i{}'\beta)(-x_i{}'\beta) = 0$$

$$\begin{aligned} V(u_i) = E(u_i^2) &= P[y_i = 1](1 - x_i{}'\beta)^2 + P[y_i = 0](-x_i{}'\beta)^2 \\ &= x_i{}'\beta(1 - x_i{}'\beta)^2 + (1 - x_i{}'\beta)(-x_i{}'\beta)^2 \\ &= x_i{}'\beta(1 - x_i{}'\beta)[(1 - x_i{}'\beta) + x_i{}'\beta] \\ &= x_i{}'\beta(1 - x_i{}'\beta) \end{aligned}$$

in which case u_i is heteroscedastic.

3. We cannot constrain $0 \leq x_i{}'\beta \leq 1$, therefore $P(y_i = 1)$ or $P(y_i = 0)$ is not bounded between zero and unity.

See Appendix A for some Stata output for an empirical example of estimating LDV models.
There are advantages to using the LPM:

1. The model is easy to estimate.

2. It is easy to interpret the coefficients.

3. It is easy to do Instrumental Variable (IV) estimation.

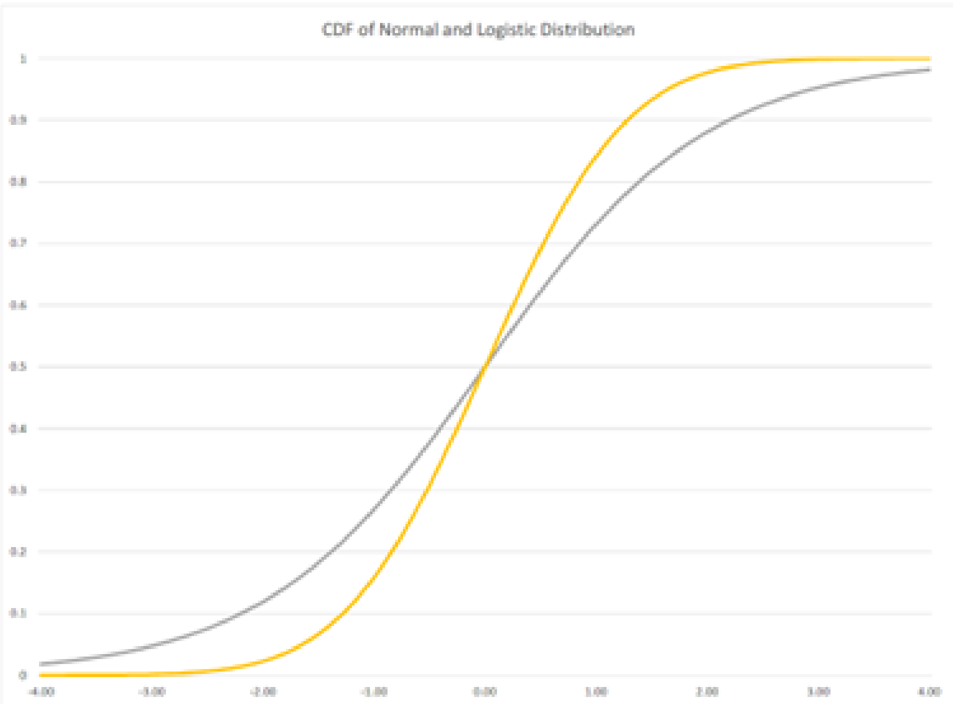
2.2 Logit Model

This has the cdf of a logistic distribution written as

$$E(y_i) = \Pr[y_i = 1] = F(x_i{}'\beta) = \frac{\exp(x_i{}'\beta)}{1 + \exp(x_i{}'\beta)} = \Lambda(x_i{}'\beta) \tag{4}$$

(see Figure 2 for an example of a Logistic Distribution (grey line), which looks like a normal distribution (yellow line), but has fatter tails i.e. more probability in the extremes).

Figure 2:



Now given n randomly selected individuals of which n_1 have $y_i = 0$ and n_2 have $y_i = 1$, and $n_1 + n_2 = n$, then the joint probability density function is written as the product of the individual density function (due to the random sample, we have independence between the outcomes implying

$P(A \cap B \cap C) = P(A).P(B).P(C))$ that is,

$$\begin{aligned} &P[y_1 = 0, \ldots, y_{n_1} = 0, y_{n_1+1} = 1, \ldots, y_n = 1] \\ &\qquad = P[y_1 = 0]P[y_2 = 0] \ldots P[y_{n_1} = 0]P[y_{n_1+1} = 1] \ldots P[y_n = 1] \\ &\qquad = \prod_{y_i=0} \overbrace{(1 - F(x_i{}'\beta))}^{n_1} \prod_{y_i=1} \overbrace{F(x_i{}'\beta)}^{n_2} \end{aligned}$$

we denote this joint density function as the likelihood function,

$$L(.) = \prod_{i=1}^{n} \{F(x_i{}'\beta)\}^{y_i} \{1 - F(x_i{}'\beta)\}^{1-y_i}$$

Taking logarithms of this expression we get the log likelihood function,

$$\ln(L(.)) = \sum_{i=1}^{n} [y_i \ln(F(x_i{}'\beta))] + [(1 - y_i) \ln(1 - F(x_i{}'\beta))]$$

that is,

$$\ln(L(.)) = \sum_{i=1}^{n} [y_i \ln(\Lambda(x_i{}'\beta))] + [(1 - y_i) \ln(1 - \Lambda(x_i{}'\beta))] \tag{5}$$

or

$$\ln(L(.)) = \sum_{i=1}^{n_2} x_i{}'\beta - \sum_{i=1}^{n} \ln(1 + e^{x_i{}'\beta})$$

and this is the expression we must maximise with respect to β. To do this we must partially differentiate this function with respect to β and set the derivatives to zero. This function is nonlinear and so there is no simple algebraic expression for the β estimates, although the solution is unique (see Appendix A for a Logit model example).

2.3 Probit Model

This has the cdf of a normal distribution written as:

$$E(y_i) = P(y_i = 1) = F(x_i{}'\beta) = \int_{-\infty}^{x_i{}'\beta/\sigma} (2\pi)^{-1/2} \exp(-z^2/2)\, dz = \Phi(x_i{}'\beta)$$

and has a distribution given by the yellow line in Figure 2.
It can easily be seen from this that we can estimate only β/σ and not β and σ separately (hence we assume that $\sigma = 1$). In this case the log-likelihood (from equation (5) for the logit model) is now written as

$$\ln(L(.)) = \sum_{i=1}^{n} [y_i \ln(\Phi(x_i{}'\beta))] + [(1 - y_i) \ln(1 - \Phi(x_i{}'\beta))] \tag{6}$$

and again this expression must be maximised with respect to β, the solution is again unique.
As the cumulative normal distribution and the logistic distribution are very close to each other, except in the tails (the logistic being much fatter, see Figure 2). In practice we get similar results from using (5) as opposed to (6) (see Appendix A for a Probit model example).

3 Interpreting coefficients

With the logit and probit models we cannot simply interpret β_j as the change in $E(y_i)$ for a unit change in x_{ji} because,

$$E(y_i) = 1.P(y_i = 1) + 0.P(y_i = 0) = F(x_i{}'\beta)$$

therefore,

$$\frac{\partial E(y_i)}{\partial x_{ji}} = \left\{ \frac{\partial (F(x_i{}'\beta))}{\partial (x_i{}'\beta)} \right\} \frac{\partial (x_i{}'\beta)}{\partial x_{ji}}$$

from the function of a function rule. Now this second term is this derivative simply β_j therefore we have

$$\frac{\partial E(y_i)}{\partial x_{ji}} = \left\{ \frac{\partial (F(x_i{}'\beta))}{\partial (x_i{}'\beta)} \right\} \beta_j \tag{7}$$

But, the derivative of $F(x_i{}'\beta)$ (cdf) with respect to $x_i{}'\beta$ yields the probability density function (pdf), that is,

$$\frac{\partial (F(x_i{}'\beta))}{\partial (x_i{}'\beta)} = f(x_i{}'\beta)$$

where $f(x_i{}'\beta)$ is the pdf.

3.1 Logit Model

For the logistic distribution the derivative of the cdf is:

$$F(z) = \frac{\exp(z)}{1 + \exp(z)} = \Lambda(z)$$

$$\frac{\partial \Lambda(z)}{\partial z} = (1 + \exp(z))^{-1} \exp(z) - (1 + \exp(z))^{-2} \exp(z) \exp(z)$$

$$\frac{\partial \Lambda(z)}{\partial (z)} = \frac{(1 + \exp(z)) \exp(z) - \exp(z)^2}{(1 + \exp(z))^2}$$

$$\frac{\partial \Lambda(z)}{\partial (z)} = \frac{\exp(z) + \exp(z)^2 - \exp(z)^2}{(1 + \exp(z))^2} = \frac{\exp(z)}{(1 + \exp(z))^2}$$

$$\frac{\partial \Lambda(z)}{\partial (z)} = \Lambda(z)[1 - \Lambda(z)] = \text{pdf} \tag{8}$$

3.2 Probit Model

For the standard normal distribution the derivative of the cdf is:

$$F(z) = \Phi(z)$$

$$\frac{\partial \Phi(z)}{\partial (z)} = (2\pi)^{-1/2} \exp \left\{ \frac{-(z)^2}{2} \right\} = \varphi(z) = \text{pdf} \tag{9}$$

In both equation (8) and equation (9) the response on $E(y_i)$ to changes in x_{ji} depends upon i and therefore it is standard practice to calculate these responses for a few representative individuals. These marginal effect calculations are designed for a continuous random variable, x_j. Suppose instead x_j is a binary dummy variable. In this cases there are two choices:

1. Use the formula in equations (8) or (9) above and this will give a good approximation to the true marginal effect.

2. Caculate this as the difference in probability of $y = 1$ evaluated at the different values of the variable, that is,

$$\frac{\partial E(y_i)}{\partial x_{ji}} = P(y_i = 1 | x_{ji} = 1, x_{1i} = \overline{x}_1, \ldots x_{ki} = \overline{x}_k) - P(y_i = 1 | x_{ji} = 0, x_{1i} = \overline{x}_1, \ldots x_{ki} = \overline{x}_k)$$

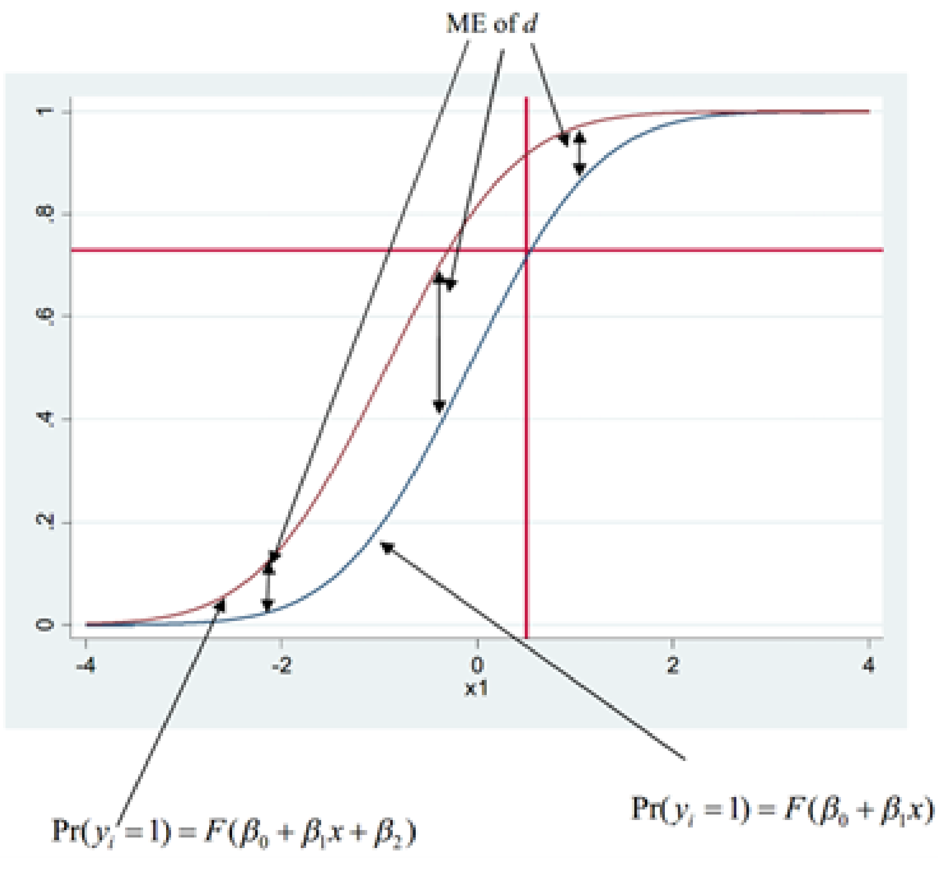
3.3 Diagrammatic exposition of ME

Consider the very simple model:

$$P(y_i = 1) = F(\beta_0 + \beta_1 x + \beta_2 d)$$

Where x is some continuous random variable and d a dummy variable. Figure 3 plots the cdf of this function over x when $d = 0$ ($\Pr(y_i = 1) = F(\beta_0 + \beta_1 x)$) and also for $d = 1(P(y_i = 1) = F(\beta_0 + \beta_1 x + \beta_2))$, where we assume that $\beta_2 > 0$. The mean of x and the corresponding mean of y are also plotted on the graph. One can clearly see that the ME of d, which is represented by the vertical distance between the two curves differs according to the value of x we show this for a small value of x, middle value of x and a large value of x.

Figure 3: ME of a dummy variable



4 Example of interpreting coefficients

Consider the case of modelling the decision to vote labour in the last election $y_i = 1$ or not $y_i = 0$. The explanatory variables are income INC, age (AGE) and union membership UNION. Estimating the model using Stata yielded the results

$$P(y_i = 1) = \Lambda[2.6 - 0.07INC_i - 0.03AGE_i + 0.4UNION_i]$$

where $UNION = 1$ if union member 0 if non − member and we know that 26% of our sample our Union members, INC (measured in £000s) varies from 6 to 56, with a mean of 21.6 and AGE varies from 21 to 74 with a mean of 46.3.
The probability of voting Labour for a union member, with income of £6,000, who is aged 26:

$$P(y_i = 1) = \Lambda(2.6 - 0.07(6) - 0.03(26) + 0.4) = \Lambda(1.8) = \frac{\exp(1.8)}{1 + \exp(1.8)} = 0.858$$

The probability of voting Labour for a non-union member, with income of £42,000, who is aged 56:

$$P(y_i = 1) = \Lambda(2.6 - 0.07(42) - 0.03(56)) = \Lambda(-2.02) = \frac{\exp(-2.02)}{1 + \exp(-2.02)} = 0.117$$

The probability of voting Labour for an average individual (even though they cannot exist):

$$P(y_i = 1) = \Lambda(2.6 - 0.07(21.6) - 0.03(46.3) + 0.4(0.26)) = \Lambda(-0.197) = \frac{\exp(-0.197)}{1 + \exp(-0.197)} = 0.451$$

To calculate the marginal effect we want:

$$\frac{\partial E(y_i)}{\partial x_{ji}} = \Lambda(\beta' X_i)[1 - \Lambda(\beta' X_i)]\beta_j$$

and we evaluate this at the average individual, and called Marginal Effect at Mean (MEM) i.e.

$$\frac{\partial E(y_i)}{\partial x_{ji}} = \Lambda(\beta' \overline{X})[1 - \Lambda(\beta' \overline{X})]\beta_j$$

The Average Marginal Effect (AME) is:

$$\frac{\partial E(y_i)}{\partial x_{ji}} = \frac{1}{n}\sum_{=1}^{n} \Lambda(\beta' X_i)[1 - \Lambda(\beta' X_i)]\beta_j$$

Table 1: Coefficient estimates and MEM from a logit model

Variable	b_j	$\overline{X}_j$	$b_j\overline{X}_j$	ME=$b_j\Lambda(.)[1-\Lambda(.)]$
Intercept	2.60	1.0	2.60	-
INC	-0.07	21.6	-1.512	-0.017
AGE	-0.03	46.3	-1.389	-0.007
UNION	0.40	0.26	0.104	0.099
$\overline{x}'b$			-0.197	
$\Lambda(\overline{x}'b)$			0.451	

As noted above for the dummy variable UNION this is a little strange as it only takes on two values 0 and 1. Therefore the ME on dummy variables more precisely as the difference in the predicted probability of voting labour for an average person who is a union member minus that average person who is not a union member, that is:

$$P(y_i = 1|UNION = 1) = \Lambda(2.6-0.07(21.6)-0.03(46.3)+0.4(1)) = \Lambda(0.099) = \frac{\exp(0.099)}{1+\exp(0.099)} = 0.525$$

$$P(y_i = 1|UNION = 0) = \Lambda(2.6 - 0.07(21.6) - 0.03(46.3)) = \Lambda(-0.301) = \frac{\exp(-0.301)}{1+\exp(-0.301)} = 0.425$$

The ME is then calculated as 0.525-0.425=0.100, a number very similar to that reported in Table 1. Suppose the coefficient estimated reported in Table 1 came from a probit model, then our marginal effects would be calculated as:

Table 2: Coefficient estimates and MEM from a probit model

Variable	b_j	$\overline{X}$	$b_j\overline{X}$	ME=$b_j\phi(.)$
Intercept	2.60	1.0	2.60	-
INC	-0.07	21.6	-1.512	-0.027
AGE	-0.03	46.3	-1.389	-0.012
UNION	0.40	0.26	0.104	0.157
$\overline{x}'b$			-0.197	
$\phi(\overline{x}'b)$			0.391	

Therefore the ME on UNION is:

$$P(y_i = 1|UNION = 1) = \Phi(2.6 - 0.07(21.6) - 0.03(46.3) + 0.4(1)) = \Phi(0.099) = 0.539$$

$$P(y_i = 1|UNION = 0) = \Phi(2.6 - 0.07(21.6) - 0.03(46.3)) = \Phi(-0.301) = 0.382$$

The ME is then calculated as 0.539-0.382=0.157, a number very similar to that reported in Table 2.

5 Hypothesis Testing

For the Logit and Probit models, we estimate the parameters, b by Maximum Likelihood Estimation (MLE). If we are correct about our assumptions of $F(x_i{}'\beta)$, then the MLE estimator, b, has nice properties:

1. The estimator is consistent (the estimator converges to the true parameter as the sample size gets big).
2. The estimator is asymptotically normally distributed.
3. The estimator is the most efficient of all alternative estimators.

To test: $H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$ we use a likelihood ratio test which is defined as:

$$LR = 2\left[\ln(L_U) - \ln(L_R)\right] \sim \chi^2_k$$

where $\ln(L_U)$ is the maximised log likelihood value of the unrestricted model and $\ln(L_R)$ is the maximised log likelihood of the restricted model under $H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$, which in our case is the model that only has an intercept.

6 Measures of goodness-of-fit

Define $\ln(L_w)$ as the maximised value of the log likelihood function. Now consider setting all slope coefficients to zero (so that we only have an intercept) the log likelihood function is $\ln(L_0)$. Then define:

$$R^2 = 1 - \frac{\ln(L_w)}{\ln(L_0)}$$

and $0 < R^2 < 1$.

Unfortunately this value has no natural interpretation. Alternatively, define:

$$\hat{y}_i = \begin{cases} 1 & if\ E(y_i) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

and construct a 2 by 2 contingency table based on the actual outcomes (y_i) and the predicted outcomes $(\hat{y}_i)$ based on $(F(x_i{}'b))$. So for example:

y_i	1	1	0	0	0	1	0	0	0	0
$F(x_i{}'b)$	0.56	0.44	0.32	0.51	0.20	0.78	0.44	0.33	0.39	0.55
$\hat{y}_i$	1	0	0	1	0	1	0	0	0	1
y_i	1	1	0	1	0	0	1	1	1	0
$F(x_i'b)$	0.55	0.65	0.22	0.52	0.38	0.41	0.65	0.71	0.56	0.51
$\hat{y}_i$	1	1	0	1	0	0	1	1	1	1

Summarising these results in a contingency table

		Actual		
		$y_i = 0$	$y_i = 1$	Total
Predicted	$P(y_i = 1) \leq 0.5$	8	1	9
	$P(y_i = 1) > 0.5$	3	8	11
	Total	11	9	20
	Correct	8	8	16

Correct(%) 80.00

If we adopted the simple (constant probability) rule

$$\hat{y}_i = \begin{cases} 1 & if\ \bar{y} > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Then we would have obtained a contingency table:

		$y_i = 0$	$y_i = 1$	Total
Predicted	$P(y_i = 1) \leq 0.5$	11	9	20
	$P(y_i = 1) > 0.5$	0	0	0
	Total	11	9	20
	Correct	11	0	11

Correct(%) 55.00

And we could calculate:

Total Gain 25.00

% Gain 45.45

Appendices

A LDV example

We are estimating a model for the labour Force Participation (denoted inlf) for women as a function of their age (age), ethnic origin (ethinc), whether they have a degree level qualification (degree) and the presence of at least 1 child under 19 (kids19). Initially we use a LPM:

Source	SS	df	MS			
				Number of obs	=	23,722
				F(5, 23716)	=	250.16
Model	182.082942	5	36.4165883	Prob > F	=	0.0000
Residual	3452.47237	23,716	.145575661	R-squared	=	0.0501
				Adj R-squared	=	0.0499
Total	3634.55531	23,721	.153220999	Root MSE	=	.38154

inlf	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
age	-.0018564	.0002312	-8.03	0.000	-.0023095	-.0014033
kids19	-.0660513	.0052189	-12.66	0.000	-.0762807	-.0558219
ethnic						
Asian	-.1615286	.0099584	-16.22	0.000	-.1810478	-.1420094
Other	-.086751	.0107173	-8.09	0.000	-.1077577	-.0657443
1.degree	.140355	.0052749	26.61	0.000	.130016	.1506941
_cons	.8890249	.0116719	76.17	0.000	.8661472	.9119025

The we estimate the model as a Logit model

```
Iteration 0:  Log likelihood = -11496.126
Iteration 1:  Log likelihood = -10898.049
Iteration 2:  Log likelihood = -10868.123
Iteration 3:  Log likelihood = -10868.045
Iteration 4:  Log likelihood = -10868.045

Logistic regression                                     Number of obs =  23,722
                                                        LR chi2(5)    = 1256.16
                                                        Prob > chi2   =  0.0000
Log likelihood = -10868.045                             Pseudo R2     =  0.0546
```

inlf	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
age	-.0149126	.0016832	-8.86	0.000	-.0182117	-.0116136
kids19	-.4898495	.0374569	-13.08	0.000	-.5632636	-.4164354
ethnic						
Asian	-.9408719	.0590988	-15.92	0.000	-1.056703	-.8250404
Other	-.5518281	.0669737	-8.24	0.000	-.6830941	-.4205622
1.degree	1.085656	.0424035	25.60	0.000	1.002546	1.168765
_cons	2.166677	.0873004	24.82	0.000	1.995571	2.337783

We can calculate marginal effects at mean (MEM)

```
Conditional marginal effects                          Number of obs = 23,722
Model VCE: OIM

Expression: Pr(inlf), predict()
dy/dx wrt:  age kids19 2.ethnic 3.ethnic 1.degree
At: age      = 43.00729 (mean)
    kids19   = .4765618 (mean)
    1.ethnic = .8752635 (mean)
    2.ethnic = .0674901 (mean)
    3.ethnic = .0572464 (mean)
    0.degree = .6490178 (mean)
    1.degree = .3509822 (mean)
```

	dy/dx	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
age	-.002115	.0002373	-8.91	0.000	-.0025801	-.0016499
kids19	-.0694735	.0052552	-13.22	0.000	-.0797735	-.0591735
ethnic						
Asian	-.1667096	.0125005	-13.34	0.000	-.1912102	-.1422091
Other	-.0877732	.0121666	-7.21	0.000	-.1116193	-.0639271
1.degree	.1394594	.0046923	29.72	0.000	.1302626	.1486562

Note: dy/dx for factor levels is the discrete change from the base level.

the Marginal Effect for degree using the difference in the probabilities

```
Predictive margins                                    Number of obs = 23,722
Model VCE: OIM

Expression: Pr(inlf), predict()
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
degree						
0	.7618003	.0034013	223.98	0.000	.7551339	.7684666
1	.9023737	.0032388	278.61	0.000	.8960257	.9087217

We can calculate these MEM by hand:

	b	$\bar{x}$	$b \times \bar{x}$	ME
age	-.01491	43.007	-0.6413	$-0.149126 \times \frac{exp(1.577839)}{(1+exp(1.577839))^2} = -0.0021$
kids19	-.4898	.4765	-0.2334	$\frac{exp(1.5778+0.23344-0.4898)}{(1+exp(1.5778+0.2334-0.4898))} - \frac{exp(1.5778+0.2334)}{(1+exp(1.5778+0.2334))} = -0.0701$
Asian	-.9408	.0675	-0.0635	$\frac{exp(1.5778+0.0635+0.0316-0.9408))}{(1+exp(1.5778+0.0635+0.0316-0.9408))} - \frac{exp(1.5778+0.0635+0.0316)}{(1+exp(1.5778+0.0635+0.0316))} = -0.1667$
Other	-.5518	.0572	-0.0316	$\frac{exp(1.5778+0.0635+0.0316-0.5518)}{(1+exp(1.5778+0.0635+0.0316-0.5518))} - \frac{exp(1.5778+0.0635+0.0316)}{(1+exp(1.5778+0.0635+0.0316))} = -0.0911$
1.degree	1.0856	.3509	0.3810	$\frac{exp(1.5778-0.3810+1.0856)}{(1+exp(1.5778-0.3810+1.0856))} - \frac{exp(1.5778-0.3810)}{(1+exp(1.5778-0.3810))} = 0.1395$
cons	2.1667	1.0000	2.1667	-
Total	-	1.5778	-	-

The Average marginal Effect (AME) is calculated as

```
Average marginal effects                              Number of obs = 23,722
Model VCE: OIM

Expression: Pr(inlf), predict()
dy/dx wrt:  age kids19 2.ethnic 3.ethnic 1.degree

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |  -.0021652   .0002437    -8.89   0.000    -.0026429   -.0016876
      kids19 |  -.0711231   .0054008   -13.17   0.000    -.0817085   -.0605377
             |
      ethnic |
      Asian  |  -.1653784   .0119983   -13.78   0.000    -.1888946   -.1418622
      Other  |  -.0887222   .0120481    -7.36   0.000     -.112336   -.0651083
             |
    1.degree |   .1405734   .0047233    29.76   0.000     .1313159     .149831
------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

We can also work out the classification table to work out the "fit" of the model:

```
Logistic model for inlf

              -------- True --------
Classified |         D            ~D  |      Total
-----------+--------------------------+-----------
     +     |     19208          4436  |      23644
     -     |        33            45  |         78
-----------+--------------------------+-----------
   Total   |     19241          4481  |      23722

Classified + if predicted Pr(D) >= .5
True D defined as inlf != 0
--------------------------------------------------
Sensitivity                     Pr( +| D)   99.83%
Specificity                     Pr( -|~D)    1.00%
Positive predictive value       Pr( D| +)   81.24%
Negative predictive value       Pr(~D| -)   57.69%
--------------------------------------------------
False + rate for true ~D        Pr( +|~D)   99.00%
False - rate for true D         Pr( -| D)    0.17%
False + rate for classified +   Pr(~D| +)   18.76%
False - rate for classified -   Pr( D| -)   42.31%
--------------------------------------------------
Correctly classified                        81.16%
--------------------------------------------------
```

This is the same equation estimated using a Probit model

```
Probit regression                                       Number of obs =  23,722
                                                        LR chi2(5)    = 1261.30
                                                        Prob > chi2   =  0.0000
Log likelihood = -10865.478                             Pseudo R2     =  0.0549

------------------------------------------------------------------------------
        inlf | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |  -.0088068    .000937    -9.40   0.000    -.0106433   -.0069703
      kids19 |  -.2831949   .0211753   -13.37   0.000    -.3246977   -.2416921
             |
      ethnic |
       Asian |  -.5425836    .035419   -15.32   0.000    -.6120035   -.4731636
       Other |  -.3163407   .0392433    -8.06   0.000    -.3932563   -.2394252
             |
    1.degree |   .5970521    .022461    26.58   0.000     .5530294    .6410749
       _cons |   1.296743   .0483977    26.79   0.000     1.201885      1.3916
------------------------------------------------------------------------------
```

Calculating the Marginal Effects at Mean (MEM) estimates

```
Conditional marginal effects                            Number of obs = 23,722
Model VCE: OIM

Expression: Pr(inlf), predict()
dy/dx wrt:  age kids19 2.ethnic 3.ethnic 1.degree
At: age      = 43.00729 (mean)
    kids19   = .4765618 (mean)
    1.ethnic = .8752635 (mean)
    2.ethnic = .0674901 (mean)
    3.ethnic = .0572464 (mean)
    0.degree = .6490178 (mean)
    1.degree = .3509822 (mean)

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         age |  -.0022632   .0002399    -9.44   0.000    -.0027333   -.0017931
      kids19 |  -.0727779   .0054098   -13.45   0.000    -.0833809   -.0621749
             |
      ethnic |
       Asian |  -.1658989   .0124395   -13.34   0.000    -.1902799    -.141518
       Other |  -.0889869   .0122635    -7.26   0.000     -.113023   -.0649508
             |
    1.degree |   .1406879   .0047276    29.76   0.000      .131422    .1499538
------------------------------------------------------------------------------
Note: dy/dx for factor levels is the discrete change from the base level.
```

Calculating the Marginal Effect for the factor variable degree using the difference in the probabilities

```
Predictive margins                                      Number of obs = 23,722
Model VCE: OIM

Expression: Pr(inlf), predict()
```

	Margin	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
degree						
0	.7620087	.0034001	224.12	0.000	.7553446	.7686727
1	.9023084	.0032361	278.83	0.000	.8959658	.908651

So calculating by hand we get:

	b	$\bar{x}$	$b \times \bar{x}$	ME
age	-.0088	43.007	-0.3788	$-0.0088 \times \phi(1.2967) = -0.0023$
kids19	-.2832	.4765	-0.1350	$\Phi(1.2967 + 0.1350 - 0.2832) - \Phi(1.2967 + 0.1350) = -0.0728$
Asian	-.5426	.0675	-0.0366	$\Phi(1.2967 + 0.0366 + 0.0181 - 0.5426) - \Phi(1.2967 + 0.0366 + 0.0181) = -0.1659$
Other	-.3163	.0572	-0.0181	$\Phi(1.2967 + 0.0366 + 0.0181 - 0.3163) - \Phi(1.2967 + 0.0366 + 0.0181) = -0.0890$
1.degree	.5971	.3509	0.2096	$\Phi(1.2967 - 0.2096 + 0.5971) - \Phi(1.2967 - 0.2096) = 0.1407$
cons	1.2967	1.0000	2.1667	-
Total	-	1.2967	-	-

and there are the Average Marginal Effects (AME)

```
Average marginal effects                                Number of obs = 23,722
Model VCE: OIM

Expression: Pr(inlf), predict()
dy/dx wrt:  age kids19 2.ethnic 3.ethnic 1.degree
```

	dy/dx	Delta-method std. err.	z	P>|z|	[95% conf.	interval]
age	-.0022495	.0002384	-9.44	0.000	-.0027167	-.0017822
kids19	-.0723353	.0053679	-13.48	0.000	-.0828561	-.0618144
ethnic						
Asian	-.1625072	.011989	-13.55	0.000	-.1860052	-.1390093
Other	-.087998	.0119894	-7.34	0.000	-.1114969	-.0644991
1.degree	.1402997	.004717	29.74	0.000	.1310545	.149545

```
Note: dy/dx for factor levels is the discrete change from the base level.
```

and measure of "fit"

```
Probit model for inlf

              -------- True --------
Classified |         D            ~D  |      Total
-----------+--------------------------+-----------
     +     |     19224          4458  |      23682
     -     |        17            23  |         40
-----------+--------------------------+-----------
   Total   |     19241          4481  |      23722

Classified + if predicted Pr(D) >= .5
True D defined as inlf != 0
--------------------------------------------------
Sensitivity                     Pr( +| D)   99.91%
Specificity                     Pr( -|~D)    0.51%
Positive predictive value       Pr( D| +)   81.18%
Negative predictive value       Pr(~D| -)   57.50%
--------------------------------------------------
False + rate for true ~D        Pr( +|~D)   99.49%
False - rate for true D         Pr( -| D)    0.09%
False + rate for classified +   Pr(~D| +)   18.82%
False - rate for classified -   Pr( D| -)   42.50%
--------------------------------------------------
Correctly classified                        81.14%
--------------------------------------------------
```

B IV with a RHS endogenous binary variable

Consider the model, given by the following two equations:

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 x_i + \varepsilon_{1i} \tag{B1}$$

$$P(D=1) = \Phi(\alpha_0 + \alpha_1 z_{1i} + \alpha_2 z_{2i}) \tag{B2}$$

where D_i is a binary (dummy) variable. One solution people use to estimate equation (B1) when you have an endogenous binary explanatory variable given by equation (B2) is:

1. Estimate equation (B2) as a Probit model and save the predicted probabilities, denoted $\hat{D}_i$
2. Estimate (B1) by OLS using $\hat{D}_i$ in place of D_i, that is estimate the model:

   $$y_i = \beta_0 + \beta_1 \hat{D}_i + \beta_2 x_i + \upsilon_i \tag{B3}$$

   where $\upsilon_i = \beta_1(D_i - \hat{D}_i) + \varepsilon_{1i}$
   OLS only work for equation (B3) if equation (B2) is estimated by OLS (i.e. a LPM) as then:
   $cov(\hat{D}_i, \upsilon_i) = \beta_1 cov(\hat{D}_i, D_i - \hat{D}_i) + cov(\hat{D}_i, \varepsilon_{1i}) = 0$
   as OLS yields residuals $((D_i - \hat{D}_i))$ which are uncorrelated with the fitted values $(\hat{D}_i)$. Whereas this would not be the case for Probit estimation (this is sometimes referred to as the Forbidden Regression problem).

Correct Solution:

1. Estimate equation (B2) as a Probit model and save the predicted probabilities, denoted $\hat{D}_i$
2. Estimate the equation: $D_i = \gamma_0 + \gamma_1 x_i + \gamma_2 \hat{D}_i + \eta_i$ by OLS and save the fitted values $\hat{\hat{D}}_i$
3. Estimate (B1) by OLS using $\hat{\hat{D}}_i$ in place of D_i, that is estimate the model:

   $$y_i = \beta_0 + \beta_1 \hat{\hat{D}}_i + \beta_2 x_i + \upsilon_i \tag{B4}$$

   And now the OLS estimates from (B4) will be consistent as this is a proper IV approach.

C Stata Commands for LDV models

```
/* This generates a variable for in labour force (excluding students and retired people */
gen inlf=(INECAC05==1 | INECAC05==2 | INECAC05==3 | INECAC05==4 | INECAC05==5)
replace inlf=. if(INECAC05==6 | INECAC05==13 | INECAC05==24 | INECAC05==34 | /*
*/ INECAC05==20 | INECAC05==31)
/* Gnerating a variable for children under 19 */
gen kids19=(HDPCH19>=1)
replace kids19=. if(HDPCH19<0)
/* Generating a variable for if you hold a degree */
gen degree=(HIQUL15D==1)
replace degree = . if(HIQUL15D<0)
/* OLS regression */
reg inlf age kids19 i.ethnic i.degree if(sex==2 & age>=22 & age<=63)
/* Summary statistics on the variables */
su inlf age kids19 i.ethnic i.degree if(e(sample))
/* Logit model */
logit inlf age kids19 i.ethnic i.degree if(sex==2 & age>=22 & age<=63)
/* Calculating MEM */
margins, dydx(*) atmeans
/* Calculating probability of in labour force for thise with and without a degree */
margins i.degree
/* Calculating AME */
margins, dydx(*)
/* Detyermining classification of the model */
estat classification
/* Probit model */
probit inlf age kids19 i.ethnic i.degree if(sex==2 & age>=22 & age<=63)
/* Calculating MEM */
margins, dydx(*) atmeans
/* Calculating probability of in labour force for thise with and without a degree */
margins i.degree
/* Calculating AME and saving the results*/
margins, dydx(*) post
/* This saves the output of the ME to a file ldv_me */
outreg2 using ldv_me, ctitle(margins) replace excel
/* Determining classification of the model */
estat classification
```

D R Commands for LDV models

```
#In R you must install the appropriate package. Having installed the package
#you must load the library associated with that package each time you enter R.
#I used the following libraries over Term 1:

library(tidyverse)
library(reshape2)
library(foreign)
library(ggplot2)
library(haven)
library(readxl)
library(Hmisc)
library(ggpubr)
library(rstatix)
library(groupedstats)
library(car)
library(lmtest)
library(olsrr)
library(estimatr)
library(tseries)
library(margins)
library(ivreg)
library(stargazer)

# This generates a variable for in labour force (excluding students and retired people ,
# Generating a variable for children under 19
# Generating a variable for if you hold a degree
qlfs18 <- qlfs18 %>%
  mutate(INECAC05_1=as.factor(INECAC05),
         HDPCH19_1=as.numeric(as.character(HDPCH19)),
         HIQUL15D_1=as.factor(HIQUL15D))
qlfs18 <- qlfs18 %>%
  mutate(inlf = ifelse(INECAC05_1==1 | INECAC05_1==2 | INECAC05_1==3 | INECAC05_1==4
                      | INECAC05_1==5, 1,
                      ifelse(INECAC05_1==6 | INECAC05_1==13 | INECAC05_1==24 | INECAC05_1==34
                      | INECAC05_1==20 | INECAC05_1==31, NA, 0)),
         kids19= ifelse(HDPCH19_1 >= 1, 1,
                       ifelse(HDPCH19_1 <0, NA, 0)),
         degree=ifelse(HIQUL15D_1==1, 1,
                      ifelse(HIQUL15D_1 == -8 | HIQUL15D_1 == -9, NA, 0))
         )
qlfs18$inlf <- factor(qlfs18$inlf, levels = c(0, 1),
                      labels = c("OLF", "ILF"))
qlfs18$kids19 <- factor(qlfs18$kids19, levels = c(0, 1),
```

```
                      labels = c("0 kids", "1+ kids"))
qlfs18$degree <- factor(qlfs18$degree, levels = c(0, 1),
                        labels = c("no degree", "degree"))

# OLS regression
model22 <- lm(inlf ~ age + kids19 + ethnic + degree,
            data = subset(qlfs18, sex=="Female" & age>=20 & age<=67))
summary(model22)
qlfs18 <- qlfs18 %>%
  mutate(age = as.numeric(as.character(age)))
qlfs18_2 <- qlfs18 %>%
  filter(sex == "Female" & age>=20 & age<=67 & !is.na(inlf) & !is.na(ethnic)
        & !is.na(degree) & !is.na(kids19))
# Logit model
model23 <- glm(inlf ~ age + kids19 + ethnic + degree, family = binomial(link = "logit"),
               data=subset(qlfs18_2, sex=="Female" & age>=20 & age<=67))
summary(model23)
# Calculating probability of INLF=1 with and without a degree
new_qlfs <- data.frame(age = mean(qlfs18_2$age), kids19 = "0 kids", ethnic = "White", degree = "no de
predict(model23,  newdata = new_qlfs, type = "response")

new_qlfs <- data.frame(age = mean(qlfs18_2$age), kids19 = "0 kids", ethnic = "White", degree = "degre
predict(model23,  newdata = new_qlfs, type = "response")
# Calculate MEM
summary(margins(model23, at = list(age=mean(qlfs18_2$age),
                                    kids19 = c("0 kids","1+ kids"),
                                    ethnic = c("White", "Asian", "Other"),
                                    degree = c("no degree", "degree")),
                variables = c("age")))

# Calculating AME
margins(model23)
# Determining classification of the model
qlfs18_2$inlflhat <- predict(model23, qlfs18_2, type="response")
qlfs18_2 <- qlfs18_2 %>%
         mutate(inlfl_1=ifelse(inlflhat>=0.5,1,0))

qlfs18_2$inlfl_1 <- factor(qlfs18_2$inlfl_1, levels = c(0, 1),
                        labels = c("PL-OLF", "PL-ILF"))
table(qlfs18_2$inlf, qlfs18_2$inlfl_1)

# Probit model
model24 <- glm(inlf ~ age + kids19 + ethnic + degree, family = binomial(link = "probit"),
               data=subset(qlfs18_2, sex=="Female" & age>=22 & age<=63))
summary(model24)
```

```
# Calculating AME
margins(model24)

# Determining classification of the model
qlfs18_2$inlfphat <- predict(model24, qlfs18_2, type="response")
qlfs18_2 <- qlfs18_2 %>%
  mutate(inlfp_1=ifelse(inlfphat>=0.5,1,0))

qlfs18_2$inlfl_1 <- factor(qlfs18_2$inlfl_1, levels = c(0, 1),
                           labels = c("PP-OLF", "PP-ILF"))
table(qlfs18_2$inlf, qlfs18_2$inlfp_1)

# Using stargazer to report the results of OLS, Probit and Logit
stargazer(model22, model23, model24,
          type='text', digits = 2,
          title = 'Table 2: LDV Models',
          style = 'qje',
          column.labels = c('OLS', 'Logit', 'Probit'),
          dep.var.labels = 'In Labour Force')

# Using stargazer to report AME but this is quite messy
model24 -> old_model
if(old_model$family$link %in% c("logit", "probit"))
  {
  require(margins)
  new_model <- old_model
  marginal_coefficients <- summary(margins(old_model))$AME
  for(i in names(marginal_coefficients)) {
    new_model$coefficients[[i]] <- marginal_coefficients[[i]]
  }
} else {print("Invalid Model")}
probit_margins <- new_model
summary(probit_margins)

model23 -> old_model
if(old_model$family$link %in% c("logit", "probit"))
{
  require(margins)
  new_model <- old_model
  marginal_coefficients <- summary(margins(old_model))$AME
  for(i in names(marginal_coefficients)) {
    new_model$coefficients[[i]] <- marginal_coefficients[[i]]
  }
} else {print("Invalid Model")}
```

```
logit_margins <- new_model
summary(logit_margins)

se_probit <- summary(margins(model24))$SE
names(se_probit) <- str_remove_all(names(se_probit), "Var_dydx_")
se_probit

se_logit <- summary(margins(model23))$SE
names(se_logit) <- str_remove_all(names(se_logit), "Var_dydx_")
se_logit

custom_se <- list(sqrt(diag(vcov(model22))),
                  se_probit, se_logit)

stargazer(model22, probit_margins, logit_margins,
          type = "text",
          # Our custom standard errors
          se = custom_se)
```

References

Dougherty, C. (2016). *Introduction to Econometrics*. OUP Catalogue. Oxford University Press.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics*. Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics*. Cengage.